

Reinforcement Learning: Key Concepts and Q-Learning

Jane

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Key Concepts in Reinforcement Learning

Term	Meaning
Agent	The learner/robot
Environment	The world (Grid) the agent interacts with
State s	A cell in the grid
Action a	Move: up, down, left, right
Reward r	Feedback for an action
Policy $\pi(s)$	The strategy (what action to take in a state)
Value Function $V(s)$	Expected total reward from state s
Q-function $Q(s, a)$	Expected reward for taking action a in state s

Mathematics Behind Reinforcement Learning I

Bellman Equation for Value Function:

$$V(s) = \sum_a \pi(a | s) \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V(s')] \quad (1)$$

Where:

$\pi(a | s)$: probability of taking action a in state s

$P(s' | s, a)$: probability of reaching state s' from s via action a

$R(s, a, s')$: reward received when transitioning from s to s' via action a

γ : discount factor, $0 < \gamma < 1$

Mathematics Behind Reinforcement Learning II

Q-Value Update Rule (Q-learning):

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2)$$

Where:

α : learning rate

r : immediate reward

$\max_{a'} Q(s', a')$: maximum estimated future reward for next state s'

Learning the Gridworld – Step-by-Step I

Let the agent be at position $(1, 1)$ in a 4×4 grid. Initially, all Q-values for each state-action pair are 0.

The agent chooses an action using the ε -greedy policy:

- With probability ε , it explores (random action).

- Otherwise, it exploits (chooses the action with the maximum Q-value).

The agent starts at position $(1, 1)$ in a 4×4 grid.

The agent selects an action using the ε -greedy policy:

- With probability ε , the agent explores.

- Otherwise, it exploits and chooses the action with the highest Q-value.

Example: The agent moves right from state $(1, 1)$ to $(1, 2)$, receives a reward of -1 , and updates its Q-value.

Goal: Optimal Policy I

The agent learns the optimal policy:

Which action to take in each state to maximize the total reward.

The learned Q-table gives the optimal policy:

$$\pi^*(s) = \arg \max_a Q(s, a) \quad (3)$$

Q-Learning Algorithm

Algorithm 1 Q-Learning Algorithm

- 1: **Input:** Learning rate α , discount factor γ , exploration rate ε , number of episodes
- 2: Initialize $Q(s, a)$ arbitrarily for all state-action pairs
- 3: **for** each episode **do**
- 4: Initialize state s
- 5: **repeat**
- 6: With probability ε , choose random action a
- 7: Else, choose action $a = \arg \max_a Q(s, a)$
- 8: Take action a , observe reward r and next state s'
- 9: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$
- 10: $s \leftarrow s'$
- 11: **until** s is terminal
- 12: **end for**

Gridworld Example

S	.	.	.
.	X	X	.
.	.	.	.
.	.	.	G

Q-Table After Training

State	Action	Q-Value
(0, 0)	right	1.809800
(0, 1)	right	3.122000
(0, 2)	right	4.580000
(0, 3)	down	6.200000
(1, 0)	down	2.066502
(1, 3)	down	8.000000
(2, 0)	right	4.256418
(2, 1)	right	6.154260
(2, 2)	down	7.999995
(2, 3)	down	10.000000

Learned Policy (Arrows in Grid)

→	→	→	↓
↓	X	X	↓
→	→	↓	↓
→	→	→	G